MILITARY SPECIFICATION

POWDER, MOLDING, PBX-9404

1. SCOPE

1.1 This specification covers one type of high explosive mixture designated as PBX–9404 Molding Powder.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR–S–366 — Sieves, Standard, for Testing Purposes

MILITARY

MIL–D–98 — Diphenylamine
MIL–H–45444 — HMX

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109 — Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

F7548644 — Box, Packing for High Explosive Assembly, Details, Packing and Marking.
F7548645 — Carton, Packing, Reuseable, Collapsible for High Explosives, Assembly Details, Packing and Marking.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

D–301–56 — ASTM Specifications and Methods of Test for Soluble Cellulose Nitrate.

(Copies of ASTM Publications may be obtained from the American Society for Testing Materials. 1916 Race Street, Philadelphia 3, Pa.).

3. REQUIREMENTS

3.1 **Material.** The molding powder shall be a mixture of HMX particles, uniformly coated by a homogenous mixture of nitrocellulose, tris-beta-chloroethyl phosphate, and diphenylamine. The constituent materials used in the manufacture of the molding powder shall comply with the following requirements:

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Constituent material	*Conforming to*
HMX	MIL–H–45444 Grade I Class A and Class B
Plastic-plasticizer	
Nitrocellulose 11.8–12.2 percent nitrogen, 18–25 centipoises viscosity	ASTM Specification D–301–56
Tris-beta-chloroethyl phosphate (CEF)	(see 6.2)
Diphenylamine	Specification MIL–D–98

3.2 Granulation. The molding powder shall be supplied in the form of free-flowing particles which shall comply with the following granulation requirements when tested as specified in 4.3.1.

U.S. Standard Sieve No.	*Cumulative percent retained*
4	5 maximum (max.)
50	95 minimum (min.)
80	98 min.

3.3 Volatiles. The volatile content shall be 0.10 percent max., when tested as specified in 4.3.2.

3.4 Composition. The analytical composition of the molding powder shall be as specified in table I when tested as specified in 4.3.3.

TABLE 1

Constituent	*Percent by weight*
HMX	94.0 ± 0.5 (a)
Plastic-Plasticizer	5.9 ± 0.5 (b)
Diphenylamine	0.10 ± 0.01

(a) The specified percentage of HMX shall consist of three parts of HMX, Class A and one part of HMX, Class B.

(b) The specified percentage of plastic-plasticizer shall consist of a mixture of 0.98 ± 0.05 parts by weight of nitrocellulose, to 1.00 part of tris-beta-chloroethyl phosphate (CEF).

3.5 Color. The color of the molding powder shall be white to light brown at the time of acceptance when determined as specified in 4.3.4.

3.6 Bulk Density. The bulk density of the molding powder shall be 0.95 grams (gm.) per cubic centimeter (cc.) min., when determined as specified in 4.3.5.

3.7 Foreign matter. The molding powder shall be free of metal inclusions, wooden particles, and other foreign matter, when determined as specified in 4.3.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system. If the contractor desires to utilize a quality assurance system which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (measuring and testing equipment) and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contrac-

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tor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that effects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 **Submission of product.** At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Quantity of product inspected.
(c) Specification number and date, together with an identification and date of changes.
(d) Results obtained for all inspection performed shall be made available upon request by the contracting officer.
(e) Certificates of analysis on all material procured directly by the contractor, when such material is controlled by Government specifications listed in any of the contractual documents, shall be made available upon request by the contracting officer.
(f) Quantity in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 **Government verification.** Using the contractor's written quality assurance procedure (see 4.1.1) this detail specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with a or b as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and depending upon the nature of the deviation, may consist of lot rejection, screening, resampling, reinstruction of the supplier's employees, or other appropriate action:

(a) Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.3.2.
(b) Verification at the point of delivery shall be in accordance with 4.1.3.2.

4.1.3.1 *Surveillance.* Surveillance shall include, but is not limited to:

(a) Observation of procedures concerning lot formation and identification.
(b) Observation of sampling procedures and application of acceptance criteria.
(c) Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.
(d) Review of procedures for control and disposition of non-conforming material.

4.1.3.2 *Product inspection.* Product inspection shall consist of Government inspection of product which has been inspected by the contractor and found to meet the quality assurance provisions of this specification to

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determine that it is of the quality required by this specification and that the contractor's records are reliable.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of a homogeneous blend of PBX–9404 from one or more batches produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions, unless otherwise specified a lot shall contain 25,000 pounds minimum. Each batch shall consist of that quantity of PBX–9404 that has been subject to the same unit chemical or physical process. Each lot shall contain:

(a) HMX Grade 1 Class A, and Grade 1 Class B, each from one lot of one manufacturer.
(b) Nitrocellulose from one homogeneous lot of one manufacturer.
(c) Tris-β-chloroethyl phosphate from one homogenous lot of one manufacturer.
(d) Diphenylamine from one homogeneous lot of one manufacturer.

4.2.2 *Examination.* Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL–STD–105. Continuous sampling plans, in accordance with Handbook ORD–M608–11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect.

4.2.2.1 Box or carton, prior to closing (see dwgs. F–7548644 or F7548645 as applicable.

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.3)*
Major:	AQL 0.25 percent		
101.	Bag liner missing or torn	Visual	01001
102.	Top of bag liner not double folded	Visual	01002
103.	Form improper	Visual	01003
Minor:	AQL 0.40 percent		
201.	Bag liner pulled too tight not allowing slack	Visual	01004

4.2.2.2 Box or carton after closing (seedwg. F–7548644 or F7548645 as applicable.

Categories	*Defects*	*Method of inspection*	*Code No.*
Major:	AQL 0.25 percent		
101.	Sealing strips or straps, as applicable, missing	Visual	02001
102.	Marking misleading or unidentifiable	Visual	02002
Minor:	AQL 1.00 percent		
201.	Sealing strips or straps, as applicable, improperly applied	Visual	02003
202.	Marking improperly applied	Visual	02004

4.2.3 *Testing.*

4.2.3.1 Determination of physical and chemical properties (see 3.2 through 3.7).

4.2.3.1.1 *Sampling by lot.* A random sample of 9 containers shall be selected from each lot of PBX–9404 in such number and amount as to insure that the sample shall be representative of the lot. When lots are comprised of 9 containers or less, each container shall be selected.

4.2.3.1.2 *Preparation of composite sample.* A 1.0 $\pm$ 0.2 ounce primary sample of the molding powder shall be removed from each of the nine containers in order to equal 9 ounces. If there are less than 9 containers, equal primary samples in sufficient quantity to equal 9 ounces, shall be removed from each container. The individual primary samples shall then be combined in order to form a homogeneous composite sample of 9 ounces. The test shall be performed as specified in 4.3. If the composite sample fails to comply with any of the requirements specified, the lot shall be rejected.

4.3 Test methods and procedures.

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4.3.1 *Granulation.* A 100 ± 1 gm. portion of the dry sample shall be placed on the specified nest of sieves, complying with Specification RR–S–366, properly superimposed and assembled with a bottom pan. The nest of sieves shall be covered and shaken for 3 minutes by means of a mechanically operated sieve shaker. The portions retained or passed by the various screens shall be weighed and the results calculated to a cumulative percentage basis. Code No. 03001

4.3.2 *Volatiles.* Approximately 10 gms. of the sample shall be weighed into a tared 100 by 15 millimeter (mm.) Petri dish. The dish shall be shaken gently to distribute the sample evenly over the bottom of the dish. The dish shall then be placed in a vacuum oven, maintained at 60 ± 2° degrees Centigrade (° C.) until the sample reaches constant weight (approximately 2 hours). The dish shall then be removed, cooled in a desiccator and weighed. The percent volatiles shall be calculated as follows: Code No. 04001

$$\text{Percent volatiles} = \frac{A \times 100}{B}$$

Where:

A = loss in weight of specimen
B = original weight of specimen

4.3.3 *Determination of composition.*

4.3.3.1 *Determination of CEF.* Code No. 05001

4.3.3.1.1 *Standardization of sodium hydroxide (NaOH) solution.* Prepare approximately 0.1 N solution of NaOH by standard techniques for the preparation of a carbonate-free base. Weight out two drops of the CEF plasticizer in current use and follow the procedures as outlined for the molding power. The CEF equivalents is the weight of CEF (gm) /volume of NaOH (ml.).

4.3.3.1.2 *Reagents and slurry.*

4.3.3.1.2.1 *Sulfuric-nitric acid solution.* Mix three parts (by volume) of concentrated sulfuric acid (specific gravity (sp. gr.) 184) to one part (by volume) of concentrated nitric acid (sp. gr. 1.42).

4.3.3.1.2.2 *Ammonium molybdate solution.* Add 65 gms. of ammonium molybdate $(NH_4)_6Mo_7O_{24} \cdot 4H_2O$), 225 gm. of ammonium nitrate (NH_4NO_3) and 15 ml. of concentrated ammonium hydroxide (NH_4OH, sp. gr. 0.90) to 600 ml. of distilled water. Heat and stir gently, filter and dilute to one liter.

4.3.3.1.2.3 *Potassium nitrate (KNO_3) wash solution.* Prepare a 1 percent solution by dissolving 10 gms. of KNO_3 per liter of distilled water.

4.3.3.1.2.4 *Filter pulp slurry.* Slurry approximately 30 gms. of filter pulp (ash free) in one liter of distilled water. This slurry can also be prepared with either analytical filter paper or filter tablets.

4.3.3.1.3 *Procedure.* By conventional quartering techniques obtain the specimen that is to be analyzed. If the agglomerate size of the specimen is greater than 16 mesh, gently crush the specimen so that all particles will pass through No. 16 sieve. (Large lumps cause excessive heat build up and could possibly fire.) The specimen size is chosen so that the CEF equivalent does not exceed 20 milligrams (mg.). Weigh 0.7 ± 0.1 gm. of the sample, to the nearest mg., into a tared flat bottom boiling flask. Add 5 ml. of a sulfuric-nitric acid mixture to the powdered specimen in the flask. Allow 5 minutes for the acid to wet the specimen completely. Attach the flask to a reflux condenser and place on the heating mantle. Start cold water circulating through the condenser. Turn power on heating mantle and adjust heat so that there is a continuous refluxing of nitric acid. Reflux 30 minutes. Remove heating mantle and allow flask to cool. Wash down the barrel of the condenser with several portions of cold distilled water. Transfer the clear solution from the flask to a 250 ml. beaker. Rinse the flask to insure complete transfer of solution from the flask. Add 5 ml. of filter pulp slurry, dilute to 125 ml. and add slowly, with stirring, 25 ml. of ammonium molybdate solution. Digest and allow to stand for 30 minutes before filtering. Place a No. 42 Whatman filter paper, or equivalent, in the slotted porcelain filter funnel of the vacuum

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filtrator apparatus. Add a small amount of filter pulp slurry and swirl funnel until paper is lightly covered with filter pulp. Turn on vacuum and remove excess water. Filter the solution through this prepared paper. Wash out beaker repeatedly with 1 percent KNO_3 solution to remove the last trace of acid occluded in the precipitate and paper. Place the paper and precipitate back into the original beaker. Add 20 ml. of distilled water and macerate with a stirring rod. Titrate slowly with standard NaOH solution with continuous maceration of the paper, until most of the yellow precipitate dissolves. Add two drops of phenolphthalein and continue titration and maceration until color of solution reaches a permanent pink. If preferred, an excess of NaOH may be added and the excess can be back titrated with a standardized hydrochloric acid (HCl) solution. Calculate the percent CEF as follows:

$$\text{Percent CEF} = \frac{(\text{NaOH CEF titer}) \times (\text{ml. NaOH}) \times 100}{\text{Sample weight (gms.)}}$$

4.3.3.2 *Determination of Nitrocellulose (NC).* Code No. 05002.

4.3.3.2.1 *Special apparatus.* Spectrophotometer: Capable of measuring the optical density of the liquid samples in the ultraviolet wave length range. Reproducibility of wave length positions must be 0.5 millimicron in the ultraviolet. Readings must be accurate to 0.5 percent transmission for 1 percent optical density.

Stray light must be less than 1.5 percent at the violet end of the spectrum. The instrument should have a sensitivity multiplier which shifts optical density readings so that readings may be made on that portion of the optical density scale which has the widest spread. The following two instruments have been used with satisfactory results:

1. Spectrophotometer, Beckman Model B
2. Spectrophotometer, Beckman Model DU

4.3.3.2.2 *Special reagents.* Solvent mixture —Mix one part (by volume) of di-n-butylamine with four parts (by volume) of dimethylformamide.

4.3.3.2.3 *Procedure.* For the determination of an absorbance versus concentration calibration curve, use a standard sample of PBX 9404 molding powder which contains exactly 3 percent NC. To prepare standardizing solution within 2.5-3.5 percent NC range, weigh three specimens of the standard material into 100 ml. volumetric flasks. Into flask No. 1 weigh 0.2083 gm; into flask No. 2 weigh 0.2500 gm; and into flask No. 3 weigh 0.2916 gm. of the standard molding powder. Based on a 0.2500 gm. specimen weight, these specimens will contain the equivalent of 2.5, 3.0 and 3.5 percent NC respectively. Add sufficient solvent mixture to dissolve the specimen, dilute to mark, and age for two hours. Measure the optical density of each standard with a spectrophotometer at a wave length of 325 millimicrons, using solvent mixture in the reference cell. Plot an absorbance versus concentration calibration curve from the standard solution data. The optical density of the molding powder solution is a function of the nitrogen content of the NC used to make the powder. Therefore, the instrument calibration must be made with a standard sample containing NC from the lot used in the manufacture of the powder under test. To avoid preparing standard samples of molding powder from each new lot of NC, a correction factor can be determined by comparing the optical densities of solutions of NC alone. Make a solution of 50 mg. of a standard NC in 500 cc. of solvent mixture and read its optical density. Prepare another solution with a sample of the NC lot used in the 9404 powder to be tested. Read the optical density of this solution. Calculate a correction factor, f, as follows:

$$\frac{D_S}{D} = f$$

Where:

D = optical density of the solution of NC in the powder to be tested

DS = optical density of the standard NC solution

To determine the NC concentration of an unknown, prepare duplicate solutions of the unknown and measure absorbance as prescribed for the standards. Determine the NC

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concentration of the unknown from the calibration curve. Correct the reading from the curve as follows:

$$C_c(f) = C$$

Where:

C_c = concentration read from curve
f = factor determined above
C = concentration of NC in unknown

Notes. The optical density of the molding powder solution increases 0.005 unit of optical density in a 24 hour period, because of a very slow nitramine reaction rather than a change in the NC. A variation in temperature of 24° C. produces an optical density change of 0.05 unit; hence minor temperature changes are insignificant. The relatively large concentration of RDX and HMX account for 0.06 optical density unit and must be taken into account if calibration is against NC alone.

4.3.3.3 *Determination of Diphenylamine (DPA).* Code No. 05003

4.3.3.3.1 *Special apparatus.* Spectrophotometer: Capable of measuring the optical density of liquid samples in the ultraviolet wave length range. Reproducibility of wave length positions must be 0.5 millimicron in the ultraviolet. Readings must be accurate to 0.5 percent transmission for 1 percent optical density. Stray light must be less than 1.5 percent at the violet end of the spectrum. The instrument should have a sensitivity multiplier which shifts optical density readings so that readings may be made on that portion of the optical density scale which has the widest spread. The Beckman Spectrophotometer, Model DU, has been found satisfactory.

4.3.3.3.2 *Procedure.* Obtain a dry specimen of about 10 gm. (weighed to the nearest mg.) by conventional quartering techniques. Grind specimens to a fine powder, place in a convenient flask and add 100 ml. of anhydrous diethylether. Swirl or shake until the DPA and CEF have gone into solution. Filter through a medium porosity fritted glass crucible. Evaporate ether, transfer to a 100 ml. volumetric flask and make up to volume with ethanol. This solution is used for determination of DPA (approximate concentration of DPA 0.01 mg./ml.). Determine the absorbance of the solution at 285 millimicron (mm.) against an ethanol-water blank. The DPA content is calculated as follows:

$$\text{Percent DPA} = \frac{KA \times 100}{W}$$

Where:

K = a calibration constant taken from the Beer's law curve of absorbance versus concentration
A = absorbance
W = weight of sample

The calibration constant is obtained by preparing solutions of known DPA content, in the presence of HMX and CEF, in a manner similar to that used for the NC determination in PBX 9404 Molding Powder and in the range of interest.

4.3.3.4 *Determination of HMX.* The percentage of HMX in the powder shall be calculated by difference as follows: Code No. 05004

Percent HMX = 100 − ((Percent CEF) + (Percent NC) + (Percent DPA) + (Percent Volatiles))

4.3.4 *Color.* The color of the molding powder shall be determined by visual inspection. Slight variation of color between particles, or marked discoloration of isolated particles shall not be considered a cause for rejection of the lot. Code No. 06001

4.3.5 *Bulk density.* The bulk density shall be determined by the following procedure: Code No. 07001

4.3.5.1 *Apparatus.* Plywood box; Inside dimensions, 4 by 4 by 4 inches; padded inside top and bottom with ¼ inch rubber foam. Hole in hinged top to fit a 250-cc. graduated cylinder.

4.3.5.2 *Procedure.* Weigh 200 gm. of powder. Pour the powder into a 250 cc. graduated cylinder. Enclose the cylinder in the box. Lift the protruding end of the cylinder until the bottom flange of the cylinder contacts the top of the box. Drop the cylinder. Again lift and drop the cylinder until 20 drops have been made, at the rate of one per second. Read the volume of the compacted pow-

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der in the cylinder. Calculate bulk density in gm. per cc., as follows:

$$\text{Bulk density} = \frac{A}{B}$$

Where:

A=weight of powder (gm.)
B=volume of compacted powder (cc.)

4.3.6 *Foreign matter.* The molding powder shall be inspected visually to assure that it is free of foreign matter. Code No. 08001

5. PREPARATION FOR DELIVERY

5.1 Packing. PBX 9404 Molding Powder shall be packed and marked in accordance with Drawings F7548644 or F7548645, as applicable. Not more than one lot shall be packed in one container. Boxes or cartons shall be packed full, to a uniform net weight not to exceed 60 pounds, as shown on the applicable drawing.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Packing requirements
(c) When required by the contractor, control data of the determination for each individual batch in the blended lot shall be furnished for the following, using the test procedures specified in 4.4.
    (1) Granulation
    (2) Volatiles
    (3) Composition
    (4) Color
    (5) Bulk Density
    (6) Foreign Matter

6.2 The tris-beta-chloroethyl phosphate (CEF) may be "celluflex CEF" a product of Celanese Corp. of America, 180 Madison Ave., New York 16, N. Y. or approved substantial equal.

6.3 Inspection code numbers. The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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